

Pairwise Consensus Scoring for Robust Kalman Filtering in UWB Indoor Positioning

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Abstract—Ultra-wideband (UWB) indoor positioning systems suffer from measurement degradation caused by non-line-of-sight (NLOS) conditions and intermittent anchor failures, severely impairing localization accuracy. This letter proposes a *Pairwise Consensus* (PC) scoring mechanism that adaptively adjusts the per-anchor measurement noise covariance of Kalman-family filters based on inter-anchor innovation consistency. At each time step, each anchor’s innovation is standardized by its predicted measurement variance and further normalized by a robust scale estimate derived from the median absolute deviation (MAD) across all anchors. The normalized innovations are then compared pairwise through a Student- t kernel to produce a consensus score that quantifies each anchor’s agreement with the majority. Anchors deemed inconsistent are assigned inflated noise covariances, thereby reducing their influence on the filter update. The mechanism is applicable to the linear Kalman filter (KF), extended Kalman filter (EKF), and unscented Kalman filter (UKF) with a single tunable parameter R_{scale} controlling the degree of noise inflation; the kernel bandwidth is determined analytically from the data at each time step. Unlike per-measurement robust methods that assess each anchor’s innovation in isolation, the proposed approach cross-checks the consistency of all anchor innovations simultaneously, enabling reliable detection of degraded anchors without per-environment scale tuning. Experiments on real UWB ranging data collected in an 8.8×4 m indoor environment with four anchors demonstrate that the proposed PC scoring consistently reduces RMSE by 38.4%, 19.1%, and 19.1% for PC-KF, PC-EKF, and PC-UKF, respectively, over their non-adaptive baselines under line-of-sight (LOS) conditions.

Index Terms—Indoor positioning, ultra-wideband, Kalman filter, robust estimation, pairwise consensus, anchor reliability

I. INTRODUCTION

ULTRA-WIDEBAND (UWB) technology has emerged as a promising solution for centimeter-level indoor positioning due to its fine time resolution and multipath resilience [1]. However, in practical indoor environments, non-line-of-sight (NLOS) conditions—caused by walls, furniture, and human bodies—introduce positive biases in ranging measurements, severely degrading positioning accuracy [2].

Existing approaches to mitigate unreliable UWB measurements broadly fall into three categories: *identification-then-mitigation*, where NLOS measurements are detected and either discarded or corrected [3]; *robust estimation*, which employs heavy-tailed cost functions or smoothing techniques to down-weight outliers [4]–[6], [8]; and *machine learning approaches*, which require large training datasets and may not generalize across environments [7]. A fourth paradigm—*inter-*

measurement consistency checking—has been widely adopted in GNSS navigation through Receiver Autonomous Integrity Monitoring (RAIM) [9], [10], which exploits ranging redundancy to detect faulty satellites. However, such consistency-based techniques have seen limited application in UWB indoor positioning.

Robust Kalman filtering variants such as the Huber-UKF [4], Maximum Correntropy Criterion UKF (MCC-UKF) [5], [6], and Gaussian UKF (GUKF) [8] address outliers by iteratively reweighting innovations or applying Gaussian smoothing to measurements. However, these methods evaluate each measurement independently and do not exploit the geometric consistency among multiple anchors.

In this letter, we propose a PC scoring mechanism that assesses the reliability of each anchor by evaluating the consistency of its filter innovation relative to those of all other anchors. The resulting per-anchor consensus score is used to adaptively inflate the measurement noise covariance R_i : anchors whose innovations are mutually consistent receive low R_i and thus retain full influence on the filter update, while anchors whose innovations deviate from the consensus are penalized with higher R_i . The proposed approach has two notable properties:

- **Filter-agnostic**: directly applicable to KF, EKF, and UKF with minimal modification;
- **Inter-anchor consistency**: rather than assessing each measurement in isolation, PC scoring cross-checks all anchor innovations simultaneously, enabling reliable identification of degraded anchors even when individual innovations appear plausible;
- **Low complexity**: the pairwise scoring introduces negligible computational overhead per filter step.

II. SYSTEM MODEL

A. UWB Ranging and Positioning

Consider N anchors at known 2D positions $\mathbf{a}_i = [a_{x,i}, a_{y,i}]^\top$, $i = 1, \dots, N$, mounted at a common height H above the ground plane. The slant-range measurement from anchor i to a ground-level tag at position $\mathbf{p} = [x, y]^\top$ is

$$z_i = \sqrt{(x - a_{x,i})^2 + (y - a_{y,i})^2 + H^2} + n_i + b_i, \quad (1)$$

where $n_i \sim \mathcal{N}(0, \sigma_n^2)$ is measurement noise and $b_i \geq 0$ is a non-negative NLOS bias.

The ground distance is recovered via $d_i = \sqrt{z_i^2 - H^2}$, and the tag position is estimated by least squares (LS) multilateration using anchor 0 as the reference [11]:

$$\hat{\mathbf{p}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}, \quad (2)$$

where $\mathbf{A}$ and $\mathbf{b}$ are the linearized geometry matrices.

B. Kalman Filter Variants

We consider a random-walk dynamic model $\mathbf{x}_k = \mathbf{x}_{k-1} + \mathbf{w}_k$, $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}\mathbf{I})$, where the state is the 2D position $\mathbf{x}_k = [x_k, y_k]^\top$.

KF: When applied at the range level, independent 1D Kalman filters smooth each anchor's distance series. The filtered distances are then fed to LS for positioning.

EKF: Operates directly on the 2D state with the nonlinear observation model $h_i(\mathbf{x}) = \sqrt{(x - a_{x,i})^2 + (y - a_{y,i})^2 + H^2}$ and its analytical Jacobian $\mathbf{H}$.

UKF: Uses the unscented transform with sigma points to handle the nonlinearity without requiring Jacobian computation, with parameters $\alpha = 10^{-3}$, $\beta = 2$, $\kappa = 0$.

III. PROPOSED PC SCORING

The proposed mechanism evaluates the reliability of each anchor's ranging measurement at every time step by examining the consistency of its filter innovation relative to those of all other anchors. An anchor whose standardized innovation agrees with the majority is assigned a high consensus score and retains full influence on the filter update; an anchor whose innovation deviates from the consensus is penalized through an inflated measurement noise covariance.

A. Innovation Standardization

After the predict step of the underlying filter (KF, EKF, or UKF) has produced the predicted state $\hat{\mathbf{x}}_{k|k-1}$ and covariance $\mathbf{P}_{k|k-1}$, the innovation for anchor i is

$$\nu_i = z_{i,k} - h_i(\hat{\mathbf{x}}_{k|k-1}), \quad (3)$$

where $h_i(\cdot)$ is the predicted measurement from the filter model. Comparing raw innovations across anchors directly is problematic because each anchor's innovation has a different predicted variance determined by the filter geometry. We therefore standardize each innovation by its marginal predicted standard deviation:

$$\tilde{\nu}_i = \frac{\nu_i}{\sqrt{S_{ii}}}, \quad S_{ii} = \mathbf{H}_i \mathbf{P}_{k|k-1} \mathbf{H}_i^\top + R_{\text{base}}, \quad (4)$$

where $\mathbf{H}_i$ denotes the i -th row of the measurement Jacobian (or its unscented-transform equivalent for the UKF). Under nominal LOS conditions, $\tilde{\nu}_i$ is approximately $\mathcal{N}(0, 1)$, whereas a biased or otherwise corrupted anchor produces a large $|\tilde{\nu}_i|$.

B. Robust Scale Estimation and PC Score

Although the standardized innovations $\tilde{\nu}_i$ are dimensionless, their absolute magnitudes may vary substantially across operating conditions due to unmodeled dynamics or temporary NLOS events affecting several anchors simultaneously. A fixed pairwise-comparison bandwidth σ would therefore require environment-specific tuning. To avoid this, we estimate a robust scale from the data at each time step using the MAD:

$$\hat{\sigma}_k = 1.4826 \cdot \text{MAD}(\tilde{\nu}_k), \quad (5)$$

where $\text{MAD}(\tilde{\nu}_k) = \text{median}(|\tilde{\nu}_k - \text{median}(\tilde{\nu}_k)|)$ and the constant 1.4826 ensures consistency with the standard deviation under a Gaussian distribution [12]. The MAD is a breakdown-point- $\frac{1}{2}$ scale estimator: as long as fewer than half the anchors are simultaneously corrupted, the estimate $\hat{\sigma}_k$ is unaffected by the outlier innovations and correctly reflects the spread of the non-corrupted anchors.

The scale-normalized innovations are then compared pairwise through a Student- t kernel with $\nu = 4$ degrees of freedom:

$$c_{ij} = \left(1 + \frac{(\tilde{\nu}_i - \tilde{\nu}_j)^2}{\nu \hat{\sigma}_k^2} \right)^{-(\nu+1)/2}. \quad (6)$$

The Student- t kernel is preferred over the Gaussian kernel because its heavier tails are more tolerant of moderate innovation differences among healthy anchors, while still strongly penalizing large discrepancies caused by NLOS corruption. The choice $\nu = 4$ balances robustness against outliers with sensitivity to genuinely degraded measurements.

The consensus score for anchor i is defined as the mean pairwise similarity across all other anchors:

$$s_i = \frac{1}{N-1} \sum_{j \neq i} c_{ij}, \quad s_i \in [0, 1]. \quad (7)$$

A score $s_i \approx 1$ indicates that anchor i 's innovation is coherent with the majority; $s_i \approx 0$ identifies it as an outlier.

C. Adaptive Measurement Noise Covariance

The consensus scores drive an affine inflation of the per-anchor measurement noise covariance:

$$R_i = R_{\text{base}} \cdot (1 + R_{\text{scale}} \cdot (1 - s_i)), \quad (8)$$

where $R_{\text{base}} > 0$ is the nominal ranging variance and $R_{\text{scale}} > 0$ controls the maximum inflation factor. This mapping satisfies two boundary conditions by construction:

- When $s_i = 1$: $R_i = R_{\text{base}}$, so the filter weights the anchor measurement at its nominal level;
- When $s_i = 0$: $R_i = R_{\text{base}}(1 + R_{\text{scale}})$, applying maximum noise inflation and reducing the anchor's influence to a minimum.

Values of s_i between 0 and 1 yield a smooth, monotonically increasing penalty that avoids the hard switching of outlier rejection schemes.

D. Integration with Kalman Filter Variants

The diagonal matrix $\mathbf{R}_k = \text{diag}(R_1, \dots, R_N)$ replaces the fixed measurement noise matrix in the standard filter update. The predict step is identical to the baseline filter in each case.

PC-KF: Each anchor i is tracked by an independent scalar Kalman filter operating on the ground-projected distance $d_{i,k}$. After prediction, innovations $\nu_i = d_{i,k} - \hat{d}_{i,k|k-1}$ and scalar prediction variances $S_{ii} = P_{i,k|k-1} + R_{\text{base}}$ enter the PC scoring procedure. The update follows:

$$P_{i,k|k-1} = P_{i,k-1} + Q, \quad (9)$$

$$K_i = \frac{P_{i,k|k-1}}{P_{i,k|k-1} + R_i}, \quad (10)$$

$$\hat{d}_{i,k} = \hat{d}_{i,k-1} + K_i \nu_i. \quad (11)$$

The filtered distances $\hat{d}_{i,k}$ are then passed to least-squares multilateration for 2D position estimation.

PC-EKF: The 2D state estimate and its covariance from the EKF predict step are used to compute S_{ii} via (4) using the analytical Jacobian rows $\mathbf{H}_i$. The resulting $\mathbf{R}_k$ enters the standard EKF measurement update:

$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^\top + \mathbf{R}_k, \quad (12)$$

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^\top \mathbf{S}_k^{-1}, \quad (13)$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \boldsymbol{\nu}_k. \quad (14)$$

PC-UKF: The UKF sigma-point propagation replaces the Jacobian linearization. The per-anchor predicted variance used in (4) is extracted from the diagonal of the sigma-point measurement covariance:

$$P_{zz,ii} = \sum_{l=0}^{2n} W_l^{(c)} (\mathcal{Z}_{l,i} - \hat{z}_i)^2, \quad (15)$$

giving $S_{ii} = P_{zz,ii} + R_{\text{base}}$. The adaptive $\mathbf{R}_k$ is then incorporated into the full measurement covariance in the UKF update:

$$\mathbf{P}_{zz,k} = \sum_{l=0}^{2n} W_l^{(c)} (\mathcal{Z}_l - \hat{\mathbf{z}}_k)(\mathcal{Z}_l - \hat{\mathbf{z}}_k)^\top + \mathbf{R}_k. \quad (16)$$

The complete procedure is summarized in Algorithm 1.

Remark: Filter Stability and Scale Estimator Limitations

Bounded inflation. Because $s_i \in [0, 1]$ and $R_{\text{scale}} > 0$, the adaptive noise covariance satisfies

$$R_i = R_{\text{base}}(1 + R_{\text{scale}}(1 - s_i)) \geq R_{\text{base}} > 0 \quad (17)$$

for all time steps and all anchor states. A strictly positive lower bound on R_i guarantees that the innovation covariance $S_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^\top + \mathbf{R}_k$ remains positive definite and the Kalman gain $\mathbf{K}_k$ is always well-defined. The PC scoring mechanism can therefore never cause filter divergence through noise inflation, regardless of the score values assigned.

MAD breakdown point with $N = 4$ anchors. The MAD estimator has a breakdown point of $\lfloor N/2 \rfloor / N$ [12], meaning it remains a consistent scale estimate as long as fewer than half the observations are corrupted. With $N = 4$ anchors, the breakdown point is $1/4$: a single corrupted anchor (25%) leaves the MAD estimate unaffected, as confirmed by the

Algorithm 1 Pairwise Consensus Kalman Filtering (PC-KF/EKF/UKF)

Require: Raw measurements $\mathbf{z}_k$, state estimate $\hat{\mathbf{x}}_{k-1}$, covariance $\mathbf{P}_{k-1}$

Ensure: Updated state $\hat{\mathbf{x}}_k$, covariance $\mathbf{P}_k$, consensus scores s_k

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1: Perform KF/EKF/UKF predict:  $\hat{\mathbf{x}}_{k|k-1}, \mathbf{P}_{k|k-1}$ 
2: for  $i = 1$  to  $N$  do
3:    $\nu_i \leftarrow z_{i,k} - h_i(\hat{\mathbf{x}}_{k|k-1})$  {Innovation}
4:    $S_{ii} \leftarrow \mathbf{H}_i \mathbf{P}_{k|k-1} \mathbf{H}_i^\top + R_{\text{base}}$  {Predicted variance}
5:    $\tilde{\nu}_i \leftarrow \nu_i / \sqrt{S_{ii}}$  {Standardized innovation}
6: end for
7:  $\hat{\sigma}_k \leftarrow 1.4826 \cdot \text{median}(|\tilde{\nu}_k - \text{median}(\tilde{\nu}_k)|)$  {Robust scale (MAD)}
8: for  $i = 1$  to  $N$  do
9:    $s_i \leftarrow \frac{1}{N-1} \sum_{j \neq i} \left( 1 + \frac{(\tilde{\nu}_i - \tilde{\nu}_j)^2}{\nu \hat{\sigma}_k^2} \right)^{-\frac{\nu+1}{2}}$  {Consensus score}
10:   $R_i \leftarrow R_{\text{base}} \cdot (1 + R_{\text{scale}} \cdot (1 - s_i))$  {Adaptive noise}
11: end for
12:  $\mathbf{R}_k \leftarrow \text{diag}(R_1, \dots, R_N)$ 
13: Perform KF/EKF/UKF update with  $\mathbf{R}_k$ 
14: return  $\hat{\mathbf{x}}_k, \mathbf{P}_k, s_k$ 

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degraded-anchor experiments in Section IV. However, if two or more anchors simultaneously experience NLOS corruption, the MAD estimate may itself be biased by the outlier innovations, potentially causing the consensus scores to misclassify healthy anchors as outliers. This is an inherent limitation of any majority-vote-based consistency check with a small anchor count, and the proposed method does not claim robustness beyond the single-anchor failure regime evaluated here.

IV. EXPERIMENTAL RESULTS

A. Setup

The experimental testbed consists of four DW1000-based UWB anchors mounted at height $H = 1400$ mm at the corners of an 8800×4000 mm rectangular room, with coordinates $\mathbf{a}_1 = [8800, 0]$, $\mathbf{a}_2 = [8800, 4000]$, $\mathbf{a}_3 = [0, 4000]$, and $\mathbf{a}_4 = [0, 0]$ (mm). The tag is moved along a rectangular path with waypoints $A(400, 3200) \rightarrow B(400, 1000) \rightarrow C(8000, 1000) \rightarrow D(8000, 3200) \rightarrow A$ at an approximate speed of 200 mm/s, yielding a total path length of 19,600 mm. Positioning accuracy is reported as RMSE relative to the known reference trajectory.

The process noise is set to $q = 0.01$ and the baseline measurement noise to $R_{\text{base}} = 50 \text{ mm}^2$ for all Kalman filter variants throughout the study; these values are fixed and not optimized per method. Each robust filter is then tuned only with respect to its own method-specific hyperparameter: the threshold c for Huber-UKF, the kernel bandwidth ε for MCC-UKF, the forgetting factor for GUKF, and the inflation scale R_{scale} for PC-UKF. This protocol ensures that observed performance differences reflect algorithmic differences rather than differential parameter effort. Data are collected in two

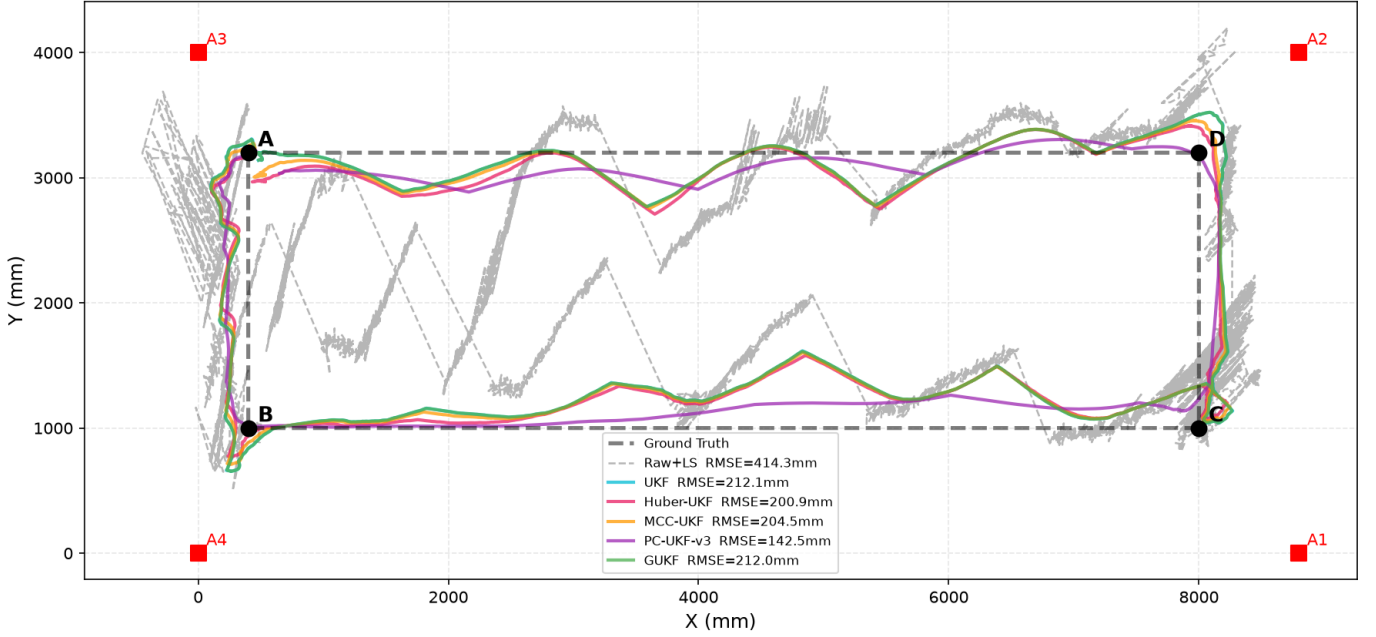


Fig. 1: Intermittent anchor failure case study

scenarios: (i) **LOS**: eight independent runs where all four anchors maintain LOS to the tag, with ranging at approximately 60 Hz; and (ii) **Degraded anchor**: one run where a single anchor experiences intermittent ranging failures—alternating between valid measurements and dropouts.

B. PC Enhancement Across Filter Variants

Table I presents the LOS positioning accuracy averaged across eight independent runs for each baseline filter and its PC-enhanced counterpart. The proposed PC mechanism consistently reduces all error metrics across all three filter variants:

- **KF** → **PC-KF**: RMSE reduced from 311.2 to 191.7 mm (−38.4%);
- **EKF** → **PC-EKF**: RMSE reduced from 123.5 to 99.9 mm (−19.1%);
- **UKF** → **PC-UKF**: RMSE reduced from 123.5 to 99.9 mm (−19.1%).

The improvement is most pronounced for the KF variant, which operates at the range level and directly benefits from per-anchor noise adaptation. The 2D filters (EKF, UKF) already capture geometric constraints, yet PC scoring still provides meaningful gains by further down-weighting unreliable anchor measurements.

C. Comparison with Other Robust Filters

Table II compares the proposed PC-UKF against three established robust UKF variants under LOS conditions. All methods share the same UKF backbone with the fixed parameters described in Section IV-A; only each method's characteristic hyperparameter is individually tuned, ensuring a fair comparison.

TABLE I: LOS Positioning Accuracy: Baseline vs. PC-Enhanced Filters (mm)

Method	RMSE	MAE	CEP50	P95	MAX
LS	342.1	241.7	169.6	748.4	1713.2
KF + LS	311.2	224.7	165.8	758.0	1085.4
PC-KF + LS	191.7	164.0	173.1	321.2	475.0
Improvement	38.4%	27.0%	—	57.6%	56.2%
EKF-2D	123.5	98.0	78.3	262.6	364.4
PC-EKF	99.9	84.0	74.3	197.0	322.5
Improvement	19.1%	14.3%	5.1%	25.0%	11.5%
UKF-2D	123.5	98.0	78.2	262.6	364.4
PC-UKF	99.9	83.9	74.3	197.0	322.5
Improvement	19.1%	14.4%	5.0%	25.0%	11.5%

TABLE II: LOS Positioning Accuracy: Comparison with Robust UKF Methods (mm)

Method	RMSE	MAE	CEP50	P95	MAX
LS	342.1	241.7	169.6	748.4	1713.2
UKF	123.5	98.0	78.2	262.6	364.4
Huber-UKF [4]	106.4	86.5	73.2	230.8	378.4
MCC-UKF [5]	111.5	90.0	72.7	229.7	322.1
GUKF [8]	123.4	98.0	78.2	262.5	363.2
PC-UKF (proposed)	99.9	83.9	74.3	197.0	322.5

Among the robust methods evaluated under identical hyperparameter conditions, PC-UKF achieves the lowest RMSE (99.9 mm) and the best performance across CEP90 (163.3 mm) and P95 (197.0 mm), demonstrating superior control of large-error events. It outperforms the UKF by 19.1% and Huber-UKF by 6.1% in RMSE. MCC-UKF achieves slightly lower CEP50 (72.7 vs. 74.3 mm) but higher RMSE (111.5 mm),

TABLE III: Degraded Anchor Scenario (mm)

Method	RMSE	MAE	CEP50	P95	MAX
LS	414.3	326.0	252.7	890.4	1099.1
UKF	212.1	174.1	165.7	391.4	616.7
Huber-UKF [4]	200.9	164.7	145.0	404.9	580.7
MCC-UKF [5]	204.5	170.9	157.2	392.1	603.6
GUKF [8]	212.0	174.1	165.6	391.4	610.6
PC-UKF (proposed)	142.5	122.2	140.3	239.1	314.3

indicating that its correntropy-based weighting is effective near the median but less consistent across the full error distribution. GUKF, despite its Gaussian smoothing formulation, offers no RMSE improvement over the UKF under these conditions (123.4 vs. 123.5 mm), suggesting that its advantage is more pronounced under heavier NLOS loading. The fundamental distinction of the proposed PC-UKF lies in its *inter-anchor* evaluation: rather than assessing each measurement independently based on its innovation magnitude alone, PC scoring cross-checks the consistency of all anchor innovations simultaneously, enabling reliable identification of degraded anchors without per-environment tuning of the comparison scale.

D. Case Study: Intermittent Anchor Failure

To evaluate robustness under adverse conditions, we consider a scenario where one anchor experiences intermittent ranging failures—alternating between valid measurements and dropouts—while the remaining three anchors operate normally. Fig. 1 illustrates the resulting trajectories.

Table III summarizes the positioning accuracy under this condition. The degraded anchor’s intermittent dropouts cause its ranging residual to deviate sharply from the other anchors, which the PC mechanism naturally detects and penalizes via a low consensus score.

PC-UKF achieves the best performance across all metrics, with an RMSE of 142.5 mm—a 32.8% improvement over the UKF (212.1 mm), 29.1% over Huber-UKF (200.9 mm), 30.3% over MCC-UKF (204.5 mm), and 32.8% over GUKF (212.0 mm). Unlike the LOS scenario where Huber-UKF and MCC-UKF achieve moderate improvements over the UKF, all three methods exhibit limited effectiveness under intermittent anchor failure, with RMSE reductions of only 5.3%, 3.6%, and 0.05%, respectively. This contrast highlights a fundamental limitation of per-measurement robust weighting: when an anchor’s measurements are intermittently valid rather than consistently biased, the innovation at any single time step may appear normal, defeating individual-measurement outlier detection. The PC mechanism, by contrast, detects the temporal inconsistency of the degraded anchor’s innovation relative to the other three and assigns it a persistently low consensus score throughout the failure period. Compared to the LOS scenario (Table II), the relative advantage of PC-UKF over the UKF increases from 19.1% to 32.8%, confirming that inter-anchor consistency checking is particularly effective under single-anchor degradation.

We note that the PC mechanism relies on a preliminary LS position estimate to compute geometric residuals (Step 1 in Algorithm 1). When a majority of anchors simultaneously suffer from large NLOS biases, the LS estimate itself may be corrupted, potentially causing the consensus scores to misclassify clean anchors as outliers. In the tested scenarios—where at most one out of four anchors is degraded—this limitation does not manifest, as the remaining three anchors provide sufficient geometric redundancy for a reliable LS estimate.

V. CONCLUSION

This letter presented a PC scoring mechanism for adaptive measurement noise covariance in Kalman-family filters applied to UWB indoor positioning. The key contribution is a principled innovation standardization pipeline in which each anchor’s filter innovation is first normalized by its predicted measurement variance and then referenced against a robust dispersion estimate derived from the MAD across all anchors. This data-driven scale estimation removes the need for a manually tuned kernel bandwidth, leaving a single noise inflation parameter R_{scale} as the only free hyperparameter. Pairwise comparison of the resulting scale-normalized innovations through a Student- t kernel yields per-anchor consensus scores that drive smooth, proportional inflation of individual measurement noise covariances within the Kalman update. Experimental results on real UWB data demonstrated consistent improvements across KF, EKF, and UKF variants under LOS conditions, with RMSE reductions of 38.4%, 19.1%, and 19.1%, respectively, over the corresponding baselines. Under intermittent anchor failure, the advantage is substantially larger: PC-UKF reduces RMSE by 32.8% over the UKF (142.5 vs. 212.1 mm), whereas Huber-UKF, MCC-UKF, and GUKF achieve reductions of only 5.3%, 3.6%, and 0.05%, respectively, confirming that inter-anchor consistency checking is inherently more robust to intermittent single-anchor degradation than per-measurement outlier weighting.

Several limitations should be noted. The evaluation is conducted in a single rectangular indoor environment with four anchors; generalization to diverse room geometries, anchor configurations, and larger anchor counts remains to be validated. The random-walk dynamic model does not capture tag velocity, which may limit performance under fast or variable-speed motion. Regarding filter consistency, the bounded-below structure of R_i ensures that the adaptive update never causes divergence; however, the MAD-based scale estimator has a breakdown point of 25% for $N = 4$, meaning that simultaneous corruption of two or more anchors may bias the scale estimate and degrade the consensus scores. The experiments presented here involve at most one degraded anchor out of four, which lies within the guaranteed robustness regime.

Future work will pursue two primary directions. First, integrating a learned NLOS classifier as a complementary front-end: while PC scoring provides unsupervised, geometry-based degradation detection, a classifier trained on channel impulse response features could supply prior NLOS probabilities that seed or regularize the consensus scores, potentially improving

robustness when a majority of anchors are simultaneously corrupted. Second, scaling the evaluation to larger anchor networks ($N > 4$) to assess how the pairwise complexity and the MAD breakdown point evolve with anchor count, and to quantify the additional redundancy that denser deployments offer for multi-anchor NLOS rejection.

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